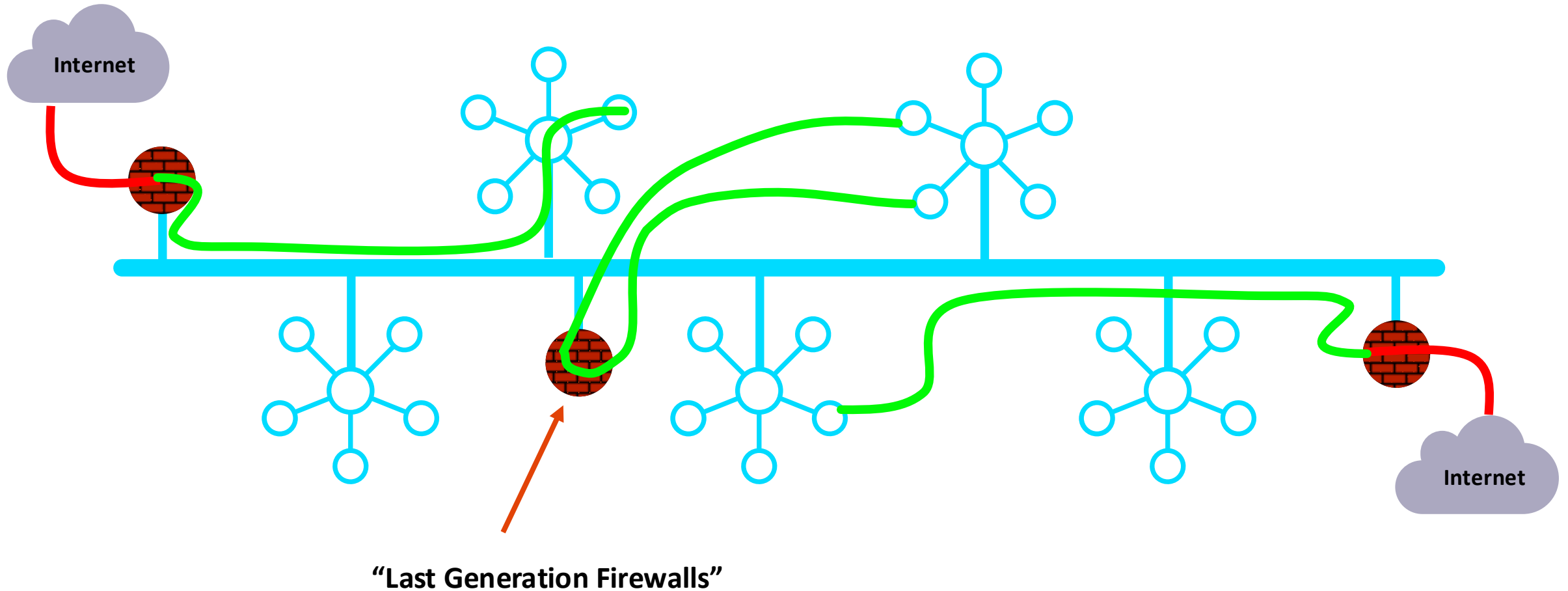




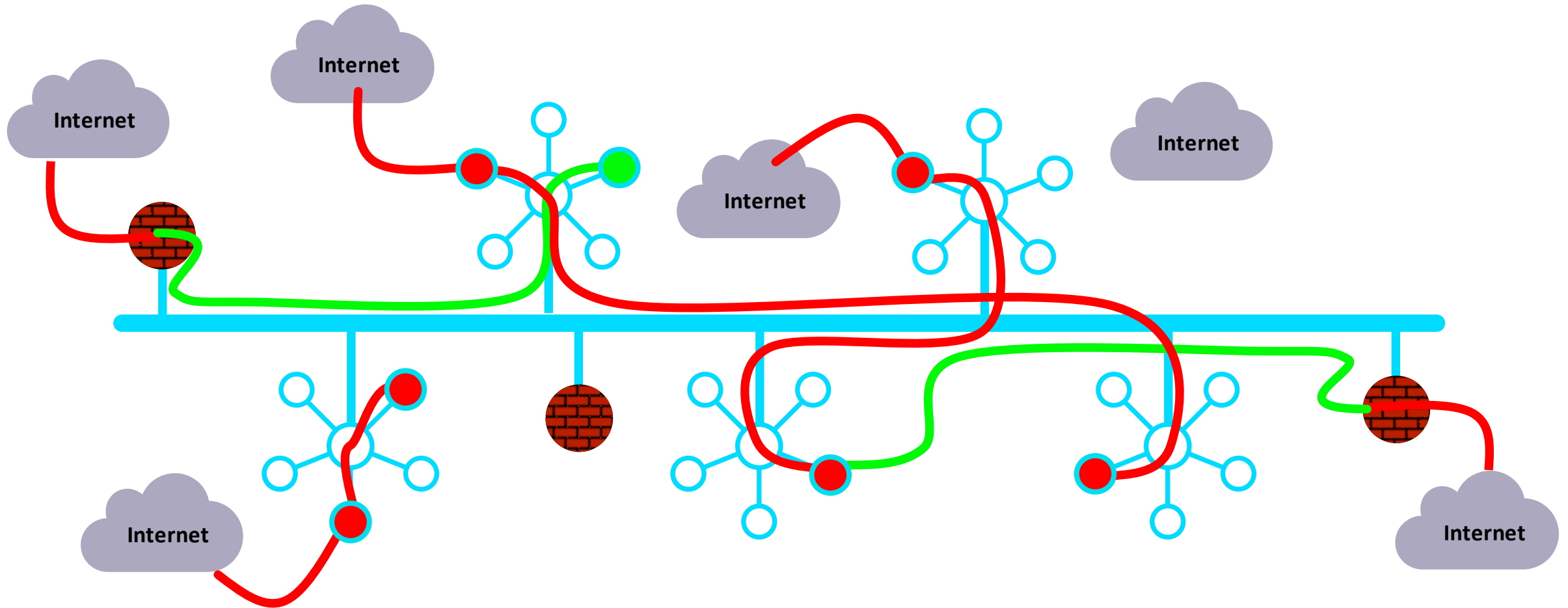
Distributed Cloud Firewall

ACE Team

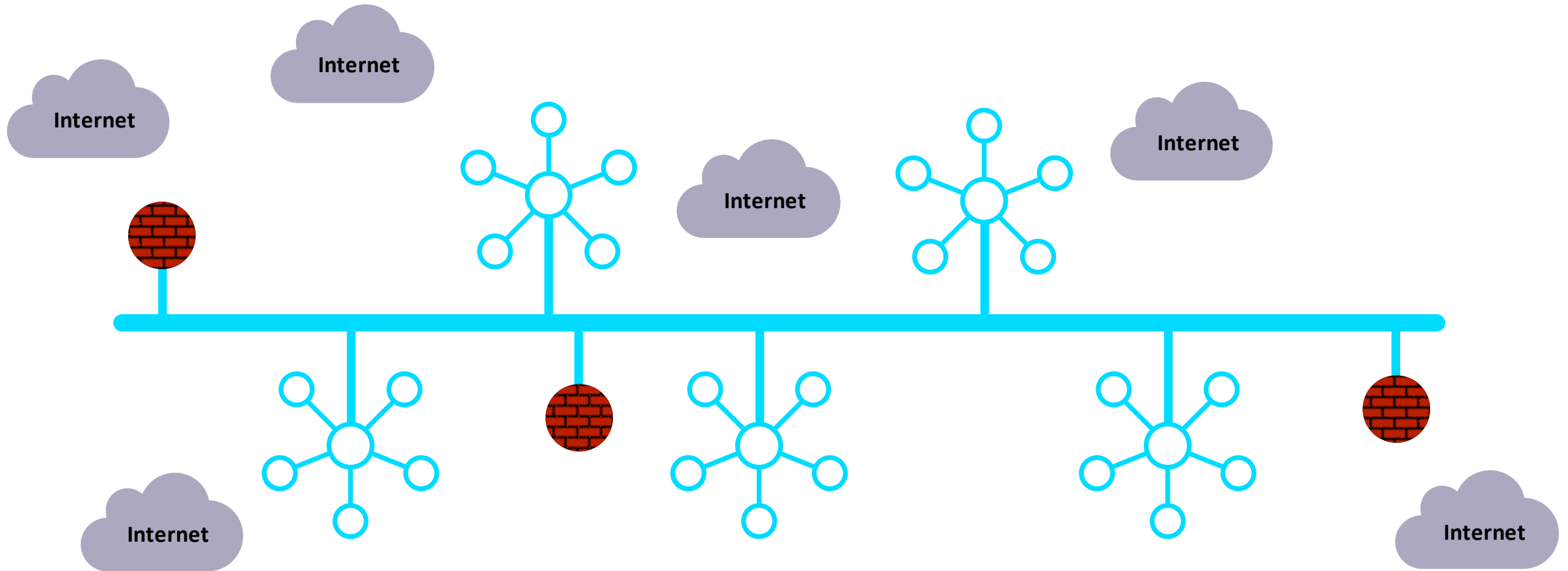
As Architected with Lift-and-Shift, Bolt-on, Data Center Era Products...



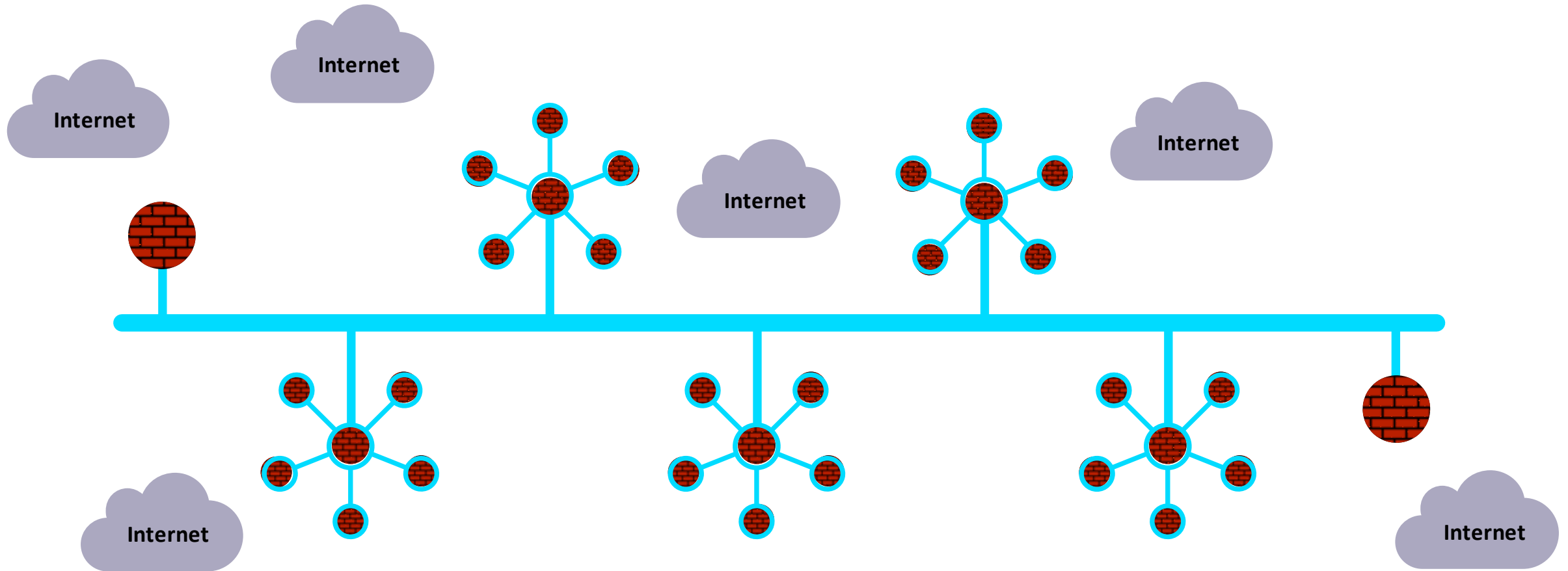
In Reality...



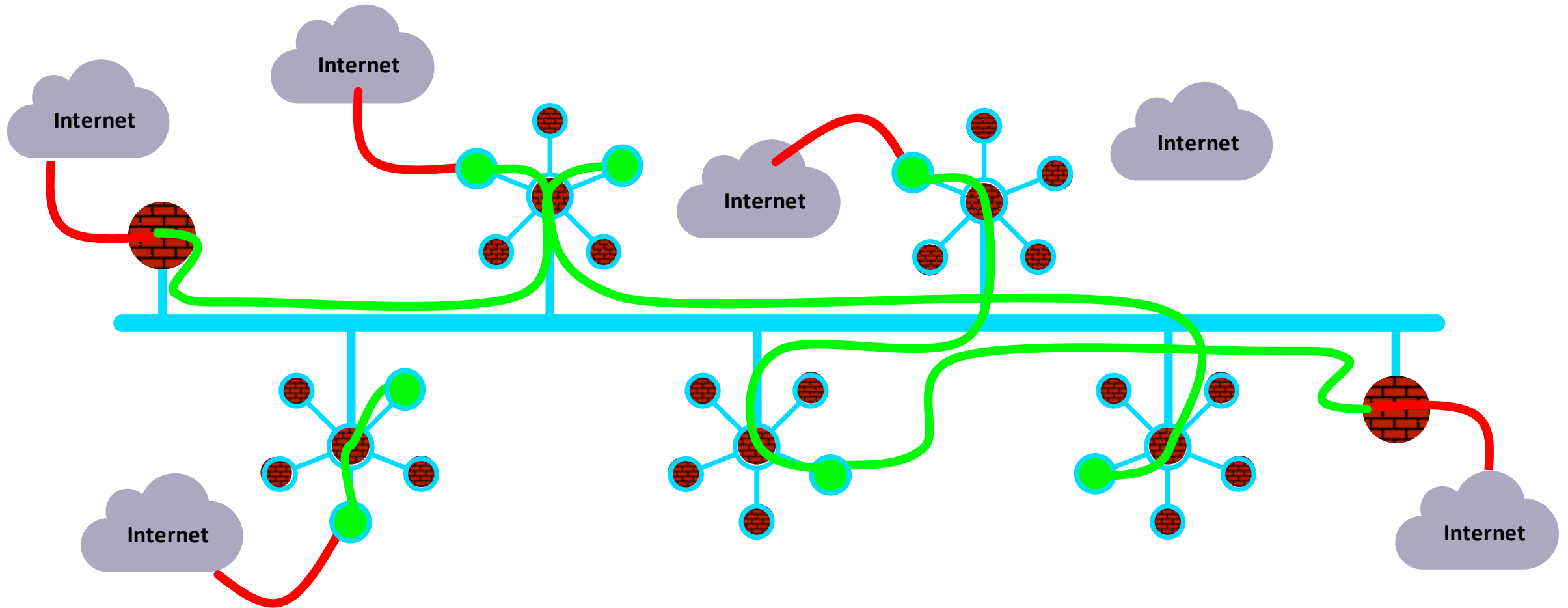
What If... the architecture was built for cloud



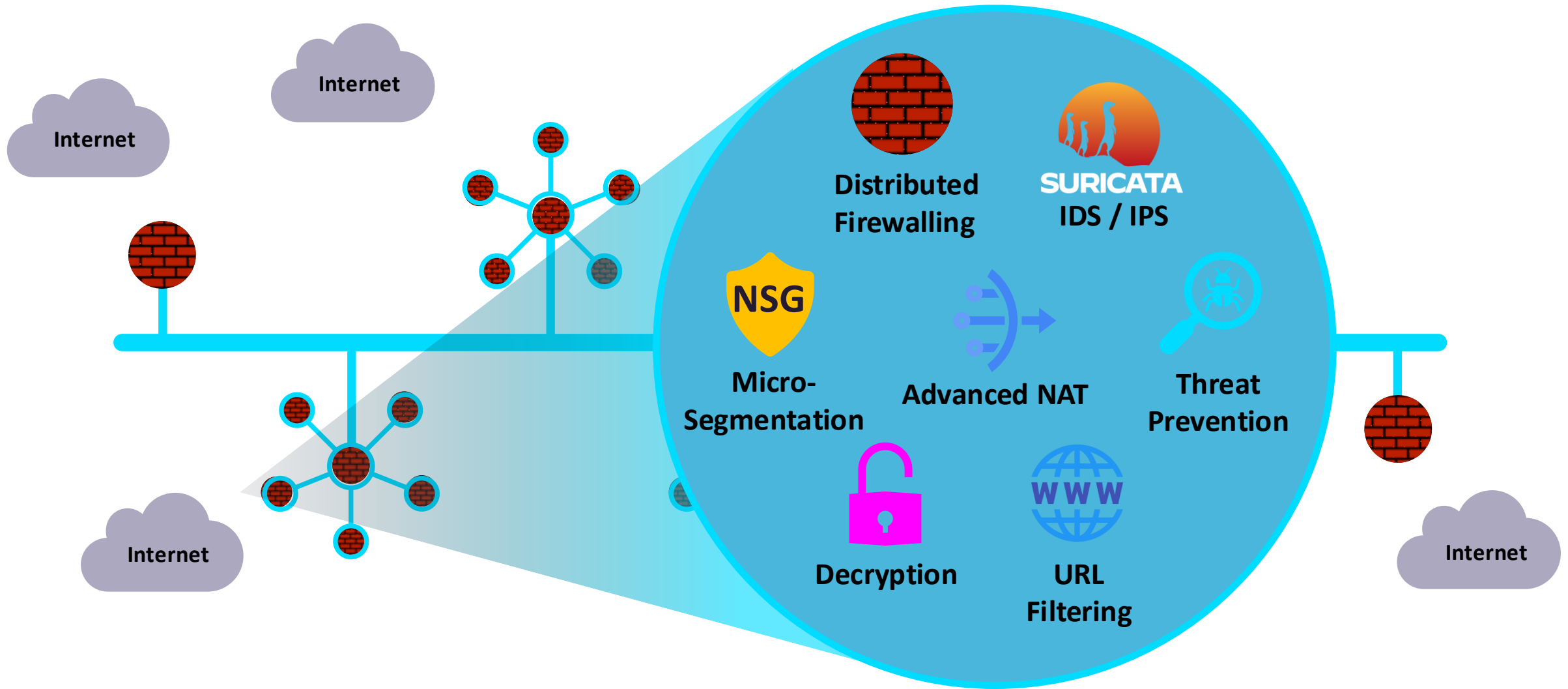
Firewalling Functions were Embedded in the Cloud Network Everywhere...



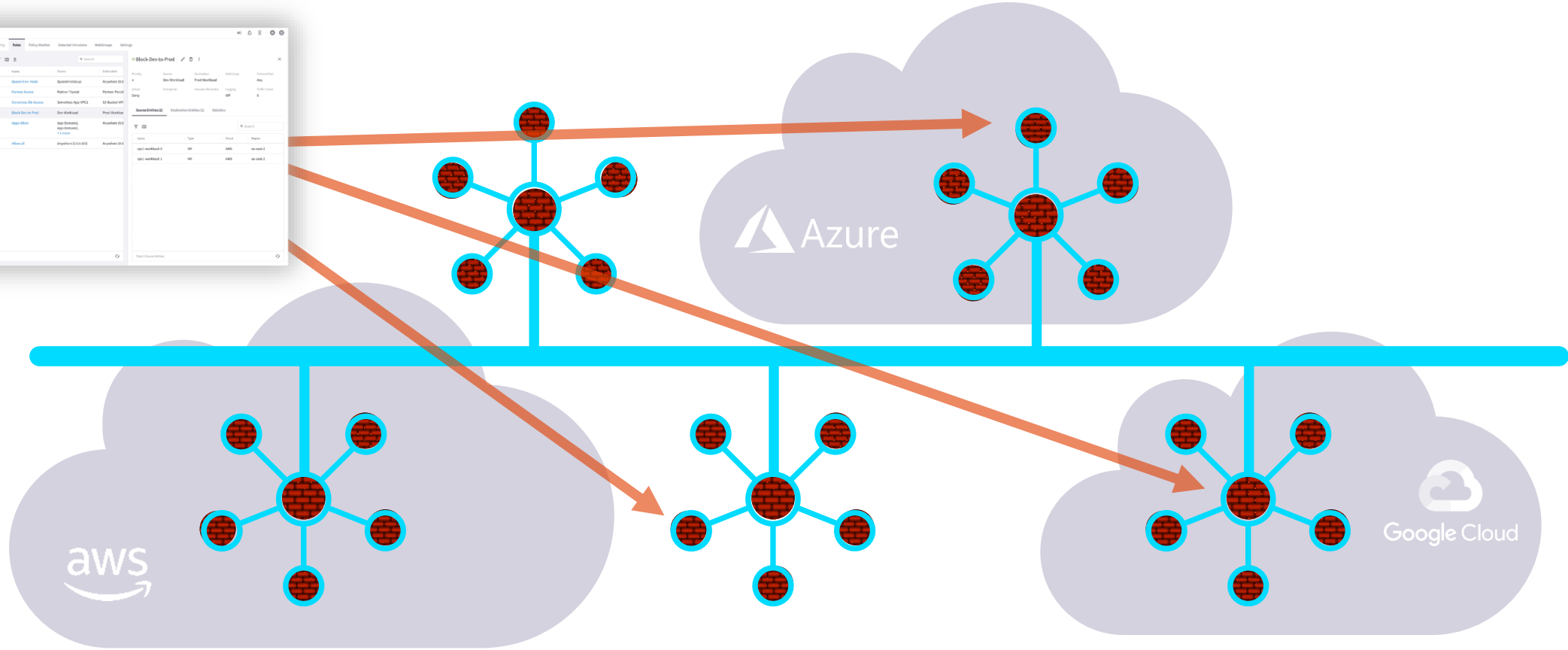
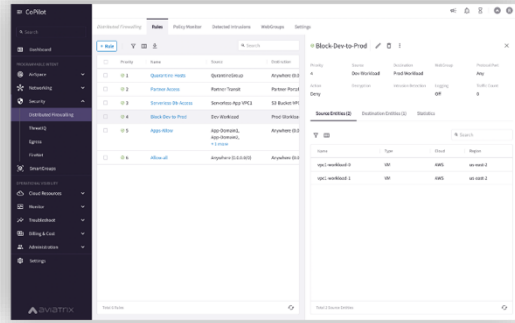
Centrally Managed, with Distributed Inspection & Enforcement...



And, What If it was more than just firewalling...

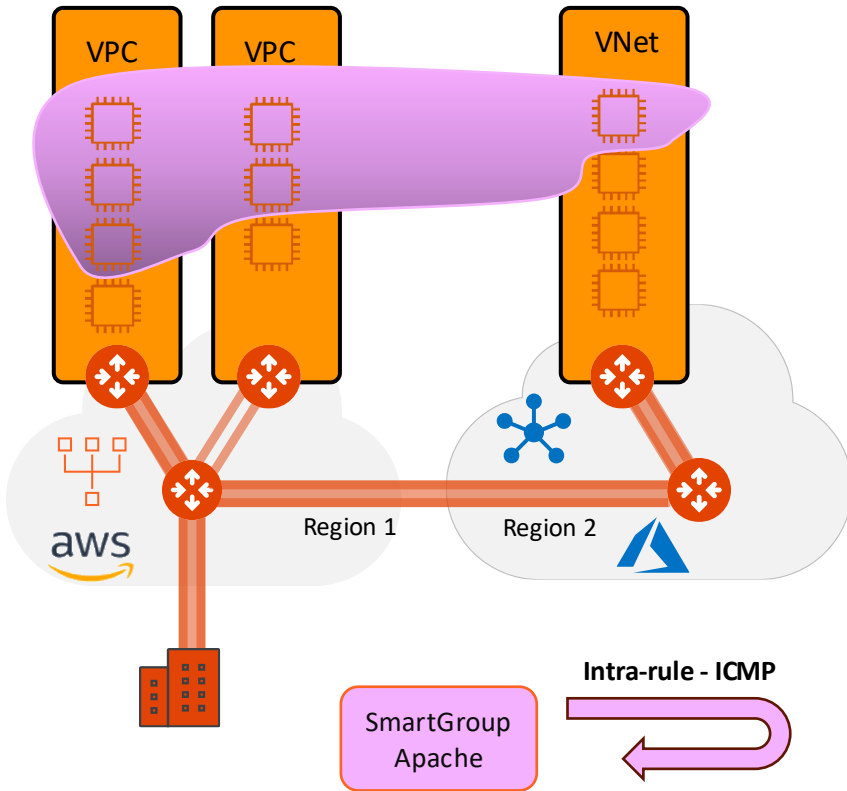


Policy Creation Looked Like One Big Firewall ... A Distributed Cloud Firewall...



Where and How Policies Are Enforced Is Abstracted...

Micro-Segmentation: SmartGroups, Intra-Rules and Inter-Rules (1)



Scenario #1

- Create a DCF rule for the APACHE SmartGroup with the following requirements:

- Permit ICMP traffic internally
- Enable the Logging feature

Create Rule

Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name
INTRA-ICMP-APACHE

Source SmartGroups
APACHE

Destination SmartGroups
APACHE

WebGroups

Protocol
ICMP

Rule Behavior

Action
Permit

SG Orchestration
On

Ensure TLS
Off

TLS Decryption
Off

Intrusion Detection (IDS)
Off

Rule Priority

Place Rule

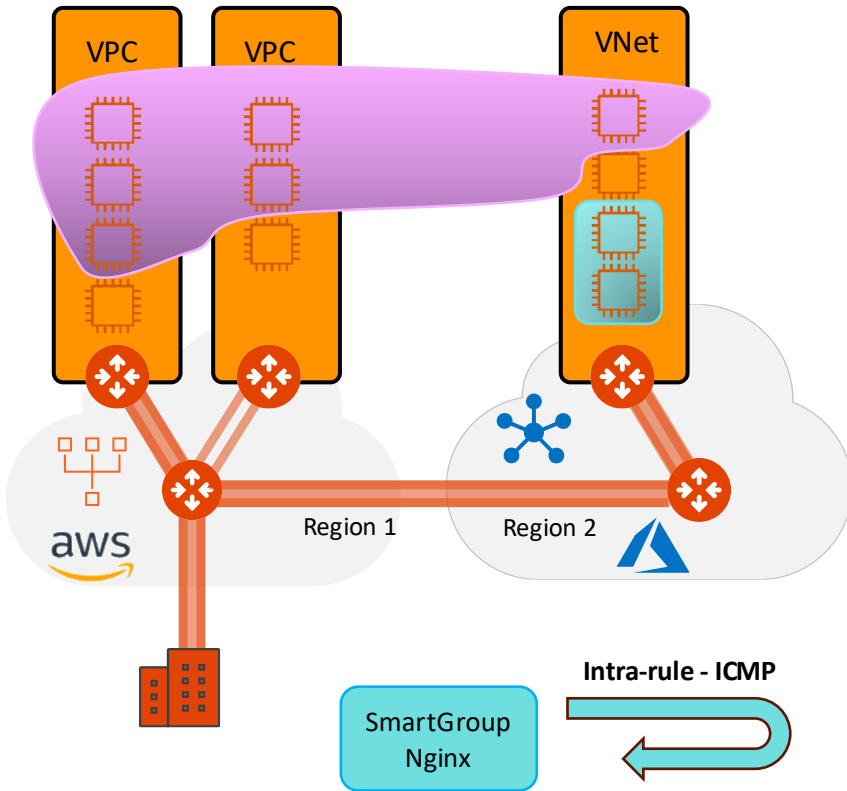
Enforcement
Logging

Cancel Save In Drafts

Intra-rule

Monitoring

Micro-Segmentation: SmartGroups, Intra-Rules and Inter-Rules (2)



Scenario #2

- ❑ Create a DCF rule for the NGINX SmartGroup with the following requirements:

- Permit ICMP traffic internally
- Enable the Logging feature

Create Rule

⚠ Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name
INTRA-ICMP-NGINX

Source SmartGroups
NGINX x

Destination SmartGroups
NGINX x

WebGroups

Protocol
ICMP

Rule Behavior
Enforcement ☐ Logging ☒

Action
Permit

SG Orchestration ☒ On

Ensure TLS ☐ Off

TLS Decryption ☐ Off

Intrusion Detection (IDS) ☐ Off

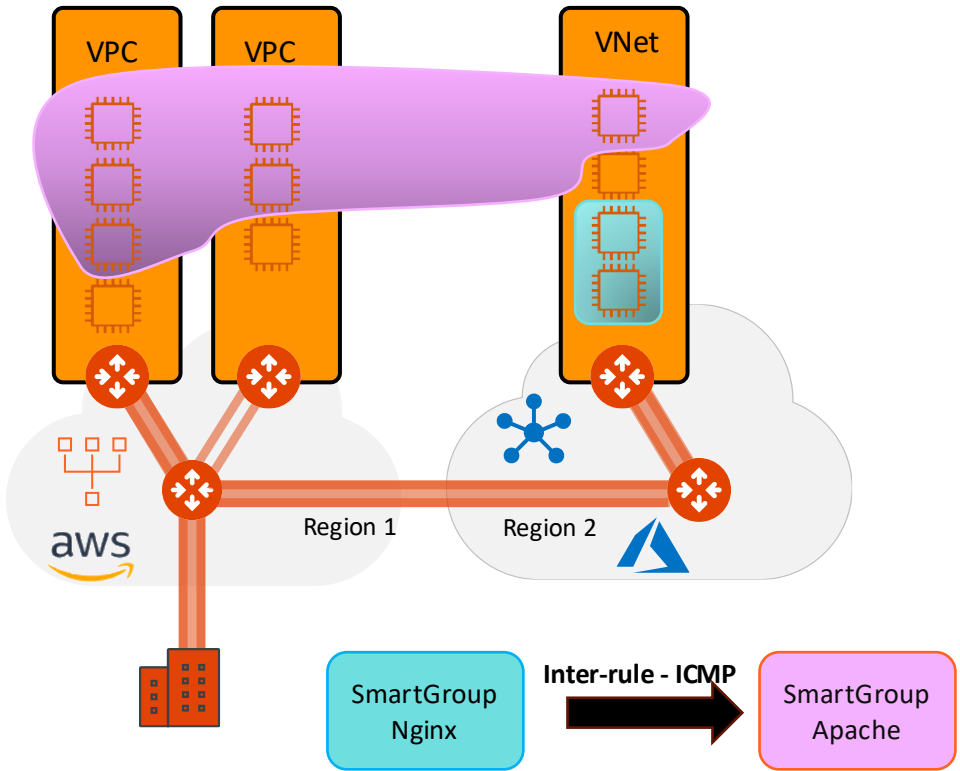
Rule Priority
Place Rule Existing Rule

Cancel Save In Drafts

Intra-rule

Monitoring

Micro-Segmentation: SmartGroups, Intra-Rules and Inter-Rules (3)



Scenario #3

- Create a DCF rule from the NGINX SmartGroup towards the APACHE SmartGroup, solely, not the inverse (NO bidirectional!), with the following requirements:

- Allow ICMP traffic between the two SGs
- Enable the Logging feature

Create Rule

Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name: INTER-ICMP-NGINX-APACHE

Source SmartGroups: NGINX

Destination SmartGroups: APACHE

WebGroups:

Protocol: ICMP

Rule Behavior: Enforcement ☒ Logging ☒

Action: Permit

SG Orchestration: ☒ On

Ensure TLS: ☐ Off

TLS Decryption: ☐ Off

Intrusion Detection (IDS): ☐ Off

Rule Priority:

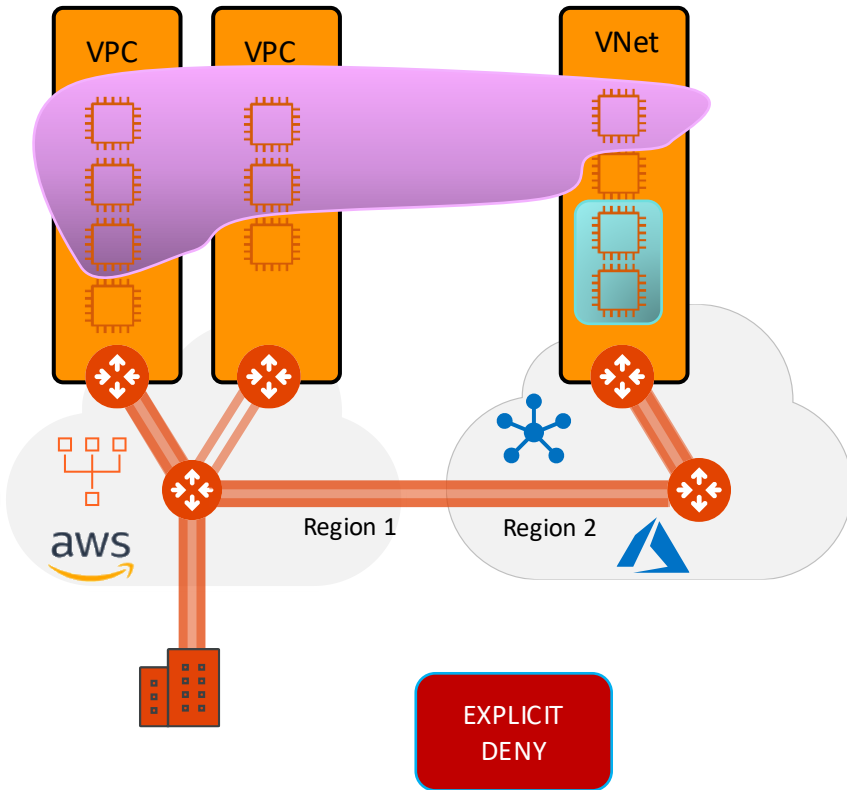
Place Rule:

Cancel Save In Drafts

Inter-rule

Monitoring

Micro-Segmentation: SmartGroups, Intra-Rules and Inter-Rules (4)



Scenario #4

- Create a DCF rule that explicitly deny any kind of traffic based on the following requirements:

- Insert the rule below the previous created rules and above the Greenfield-Rule
- Enable the Logging feature

Create Rule

Rules will be applied only on AWS, AWS Gov, ARM, ARM Gov, and GCP

Name: EXPLICIT-DENY-RULE

Source SmartGroups: Anywhere (0.0.0.0/0)

Destination SmartGroups: Anywhere (0.0.0.0/0)

WebGroups:

Protocol: Any

Port: All

Specify multiple ports (e.g. 80) and/or port ranges (e.g. 80-8080)

Rule Behavior: Enforcement ☒ Logging ☒

Action: Deny

SG Orchestration: Off

TLS Decryption: Off

Rule Priority:

Place Rule: Above

Existing Rule: Greenfield-Rule

Cancel Save In Drafts

Monitoring

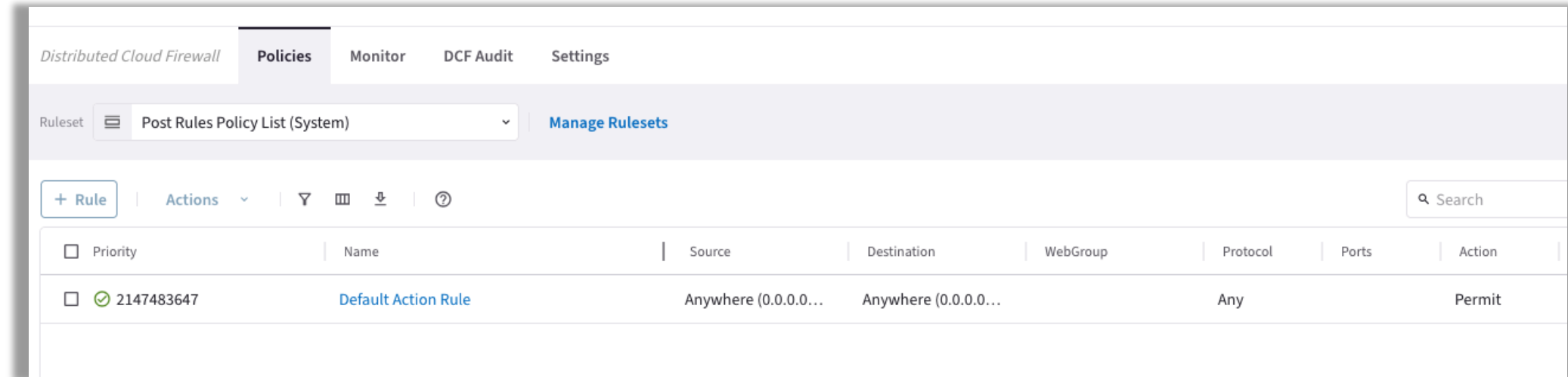
Rule Position



Tools for troubleshooting Distributed Cloud Firewall

Rulesets – Post Rules Policy List (System)

- When you enable the Distributed Cloud Firewall (DCF), the system automatically installs a **Default Action Rule** (Any-Any-Any: Permit) within the **Post Rules Policy List**.
- **Understanding the Default Action Rule:** This is a mandatory system rule that cannot be deleted. It functions as the final policy in the evaluation chain, governing any traffic not explicitly addressed by your user-defined rules.
- **Enforce Zero Trust:** To align with Zero Trust principles, you must edit this rule immediately after creation. Change the action from **Permit** to **Deny** to ensure that traffic is blocked by default if it does not match a specific allow rule.

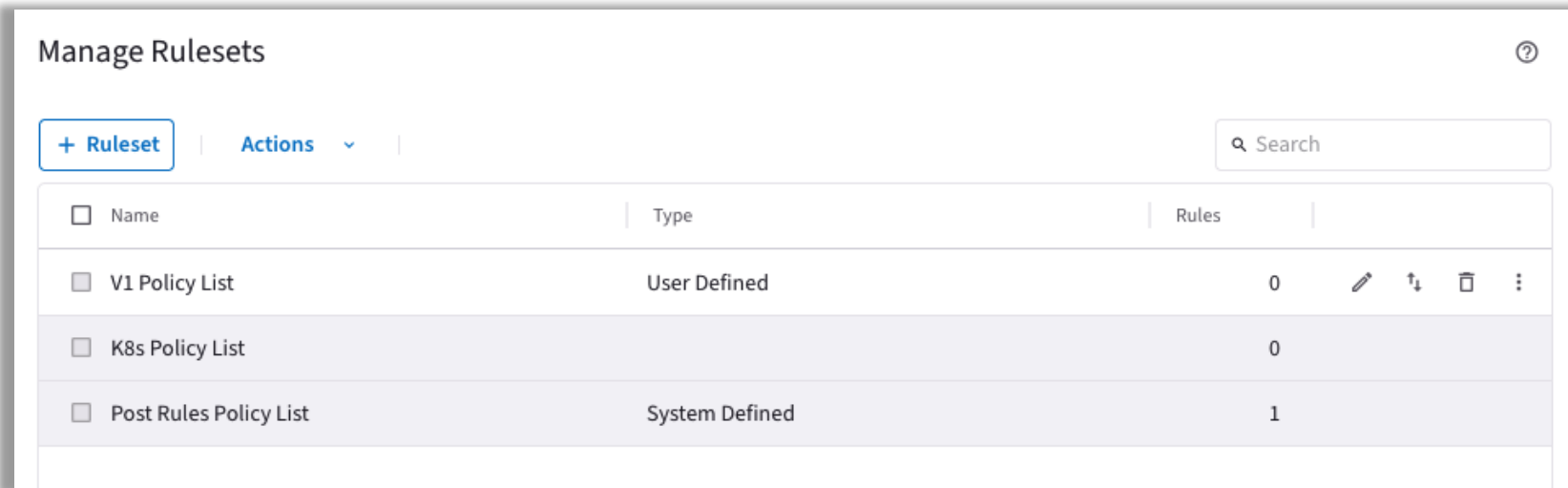


The screenshot shows the 'Policies' tab in the 'Distributed Cloud Firewall' interface. A dropdown menu shows 'Post Rules Policy List (System)' and a 'Manage Rulesets' link. Below is a table of rules with one rule visible: 'Default Action Rule' with priority 2147483647, source 'Anywhere (0.0.0.0...)', destination 'Anywhere (0.0.0.0...)', protocol 'Any', and action 'Permit'.

Distributed Cloud Firewall								Policies	Monitor	DCF Audit	Settings
Ruleset	Post Rules Policy List (System)							Manage Rulesets			
+ Rule Actions Filter Grid Download Help Search											
☐	Priority	Name	Source	Destination	WebGroup	Protocol	Ports	Action			
☐	2147483647	Default Action Rule	Anywhere (0.0.0.0...	Anywhere (0.0.0.0...		Any		Permit			

Rulesets – V1 Policy List

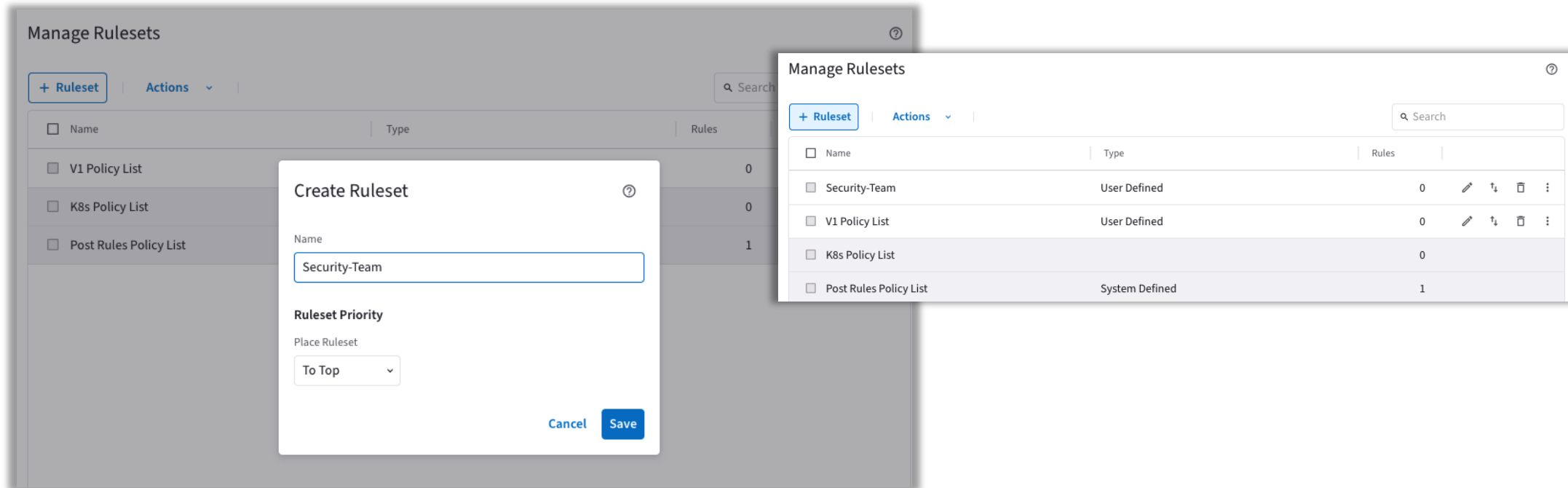
- The **V1 Policy List** is the centralized workspace where you define and manage your custom security policies. Because this list dictates the order of operations for your firewall, effective collaboration is essential.
- **Foster Cross-Team Collaboration:** Engage key stakeholders across your organization to align on security requirements and operational needs.
- **Establish Rule Precedence:** Collectively determine the hierarchy of your rulesets. This ensures that the most critical, high-priority security policies are evaluated first, maintaining a robust and predictable security posture.

The screenshot shows a web interface titled "Manage Rulesets" with a help icon. It features a "+ Ruleset" button, an "Actions" dropdown, and a search bar. Below is a table with columns for Name, Type, and Rules. The table lists three policy lists: "V1 Policy List" (User Defined, 0 rules), "K8s Policy List" (0 rules), and "Post Rules Policy List" (System Defined, 1 rule). Each row has a checkbox and a set of action icons (edit, move, delete, etc.).

☐ Name	Type	Rules	
☐ V1 Policy List	User Defined	0	   
☐ K8s Policy List		0	
☐ Post Rules Policy List	System Defined	1	

Creating a Distributed Cloud Firewall Ruleset

- **How to Create a Ruleset**
- *Follow these steps to structure your security policy hierarchy:*
- Go to **Security > Distributed Cloud Firewall > Policies** and click **Manage Rulesets**.
- **Initialize Creation:** In the *Manage Rulesets* dialog, click the **+ Ruleset** button to open the configuration window.
- **Configure Your Ruleset:** Provide the necessary details for your new ruleset

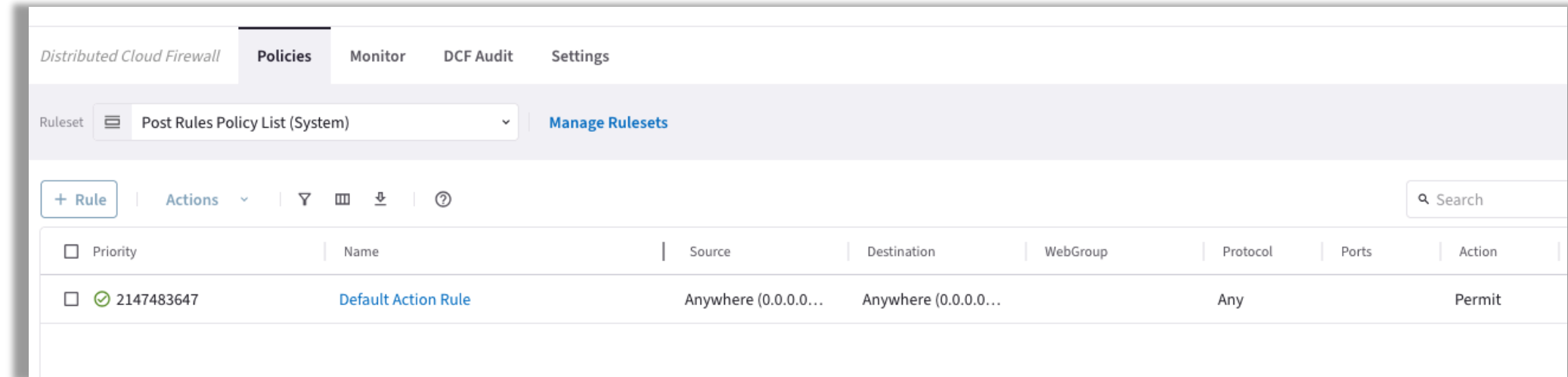


The screenshot displays the 'Manage Rulesets' interface. On the left, a list of rulesets is shown: V1 Policy List, K8s Policy List, and Post Rules Policy List. A '+ Ruleset' button is visible. In the center, a 'Create Ruleset' dialog is open, showing the 'Name' field with 'Security-Team' and the 'Ruleset Priority' dropdown set to 'To Top'. On the right, a 'Manage Rulesets' table is shown with the following data:

Name	Type	Rules	
Security-Team	User Defined	0	
V1 Policy List	User Defined	0	
K8s Policy List		0	
Post Rules Policy List	System Defined	1	

Rulesets – Post Rules Policy List (System)

- When you enable the Distributed Cloud Firewall (DCF), the system automatically installs a **Default Action Rule** (Any-Any-Any: Permit) within the **Post Rules Policy List**.
- **Understanding the Default Action Rule:** This is a mandatory system rule that cannot be deleted. It functions as the final policy in the evaluation chain, governing any traffic not explicitly addressed by your user-defined rules.
- **Enforce Zero Trust:** To align with Zero Trust principles, you must edit this rule immediately after creation. Change the action from **Permit** to **Deny** to ensure that traffic is blocked by default if it does not match a specific allow rule.



The screenshot shows the 'Policies' tab in the Distributed Cloud Firewall interface. The 'Ruleset' dropdown is set to 'Post Rules Policy List (System)'. Below the header, there is a table of rules. The first rule is the 'Default Action Rule' with priority 2147483647, source 'Anywhere (0.0.0.0...)', destination 'Anywhere (0.0.0.0...)', protocol 'Any', and action 'Permit'.

Priority	Name	Source	Destination	WebGroup	Protocol	Ports	Action
2147483647	Default Action Rule	Anywhere (0.0.0.0...)	Anywhere (0.0.0.0...)		Any		Permit

Enforcement on Clouds



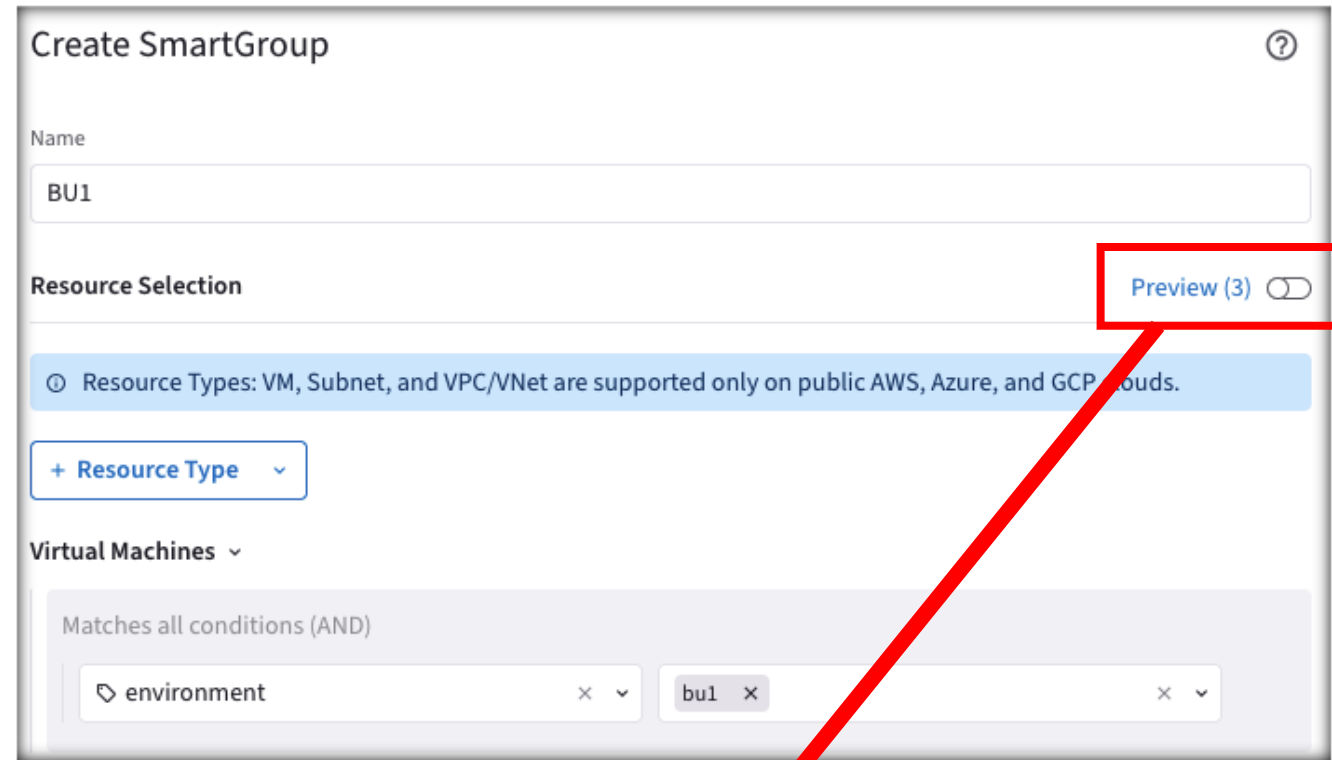
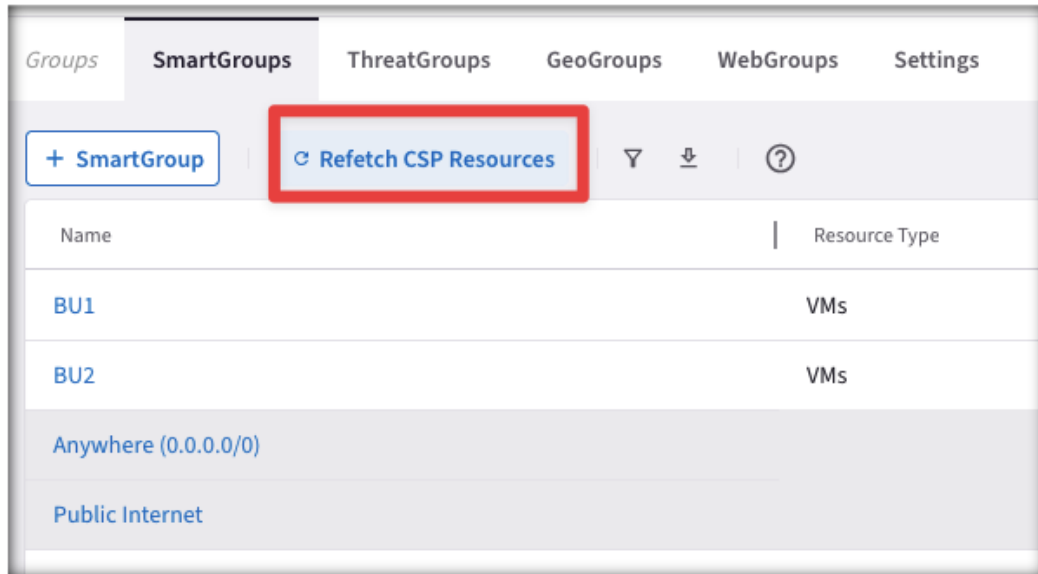
- **Advanced Enforcement Control**
- Although enforcement is applied by default to the relevant gateways within the CNSF, you have the flexibility to further refine where your policies are enacted. Through the **Manage Cloud Execution** settings, you can define specific constraints to dictate exactly which Cloud Service Provider (CSP) environments should enforce your policies.

The screenshot shows the 'Settings' tab of the Distributed Cloud Firewall interface. The 'Enforcement on Clouds' section is highlighted with a red box. The modal 'Manage Enforcement on Clouds' is open, displaying a table of cloud providers and their enforcement status. A yellow warning banner at the top of the modal states: 'Enforcement on Clouds for DCF is in Preview. Preview features are not safe for deployment in production environments. Learn More'. The table lists 10 cloud providers, with the first six having enforcement turned 'On' and the last four turned 'Off'. A 'Total 10 Clouds' summary is at the bottom of the table, along with a refresh icon and the text 'Refreshed a minute ago'. 'Cancel' and 'Save' buttons are at the bottom right of the modal.

Name	Cloud Accounts	Enforcement
AWS	1	On
Azure ARM	1	On
GCP	1	On
AWS GovCloud		On
Azure Government		On
AWS China		Off
ARM China		Off
OCI		Off
Alibaba Cloud		Off
Edge (Hybrid Cloud)	1	Off
Total 10 Clouds		Refreshed a minute ago

Creation of the SmartGroup: the right matching criteria dilemma

- Choose the right matching criteria for resources that you want to see assigned to a specific SmartGroup:
- Use the **Preview Resources** toggle switch to verify the selected resources that have been mapped to the Smart Group
- Use the On-Demand **Refetch CSP Resources** button to retrieve the most recent inventory

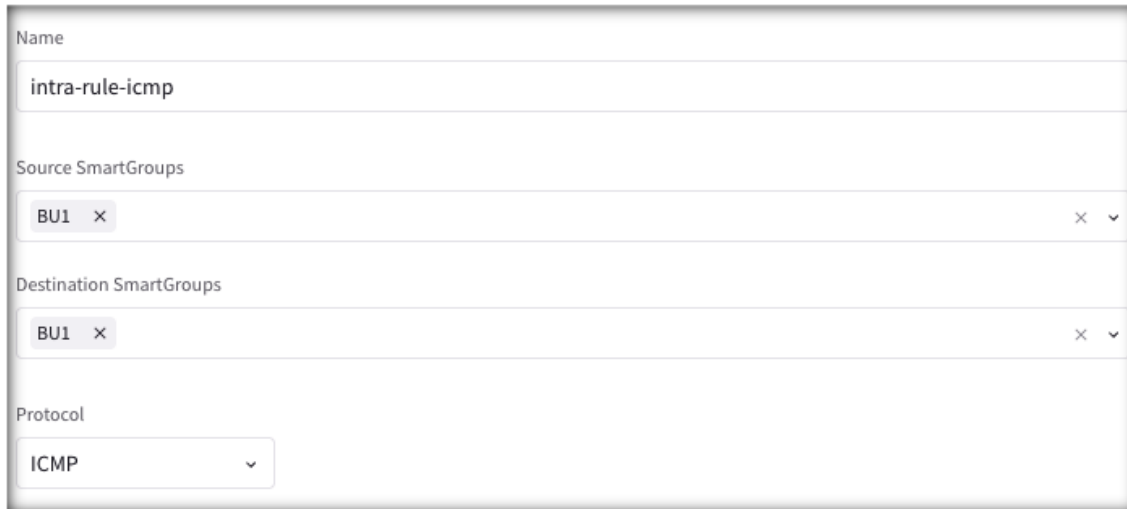


Name	Type	Cloud	Region
ace-aws-eu-west-1-spoke1...	VM	AWS	eu-west-1
ace-azure-east-us-spoke1...	VM	Azure ARM	eastus
ace-gcp-us-east1-spoke1-b...	VM	GCP	us-east1

Creation of the DCF Rules: intra-rule vs. inter-rule

1) **Intra-rule** will affect the traffic WITHIN a Smart Group

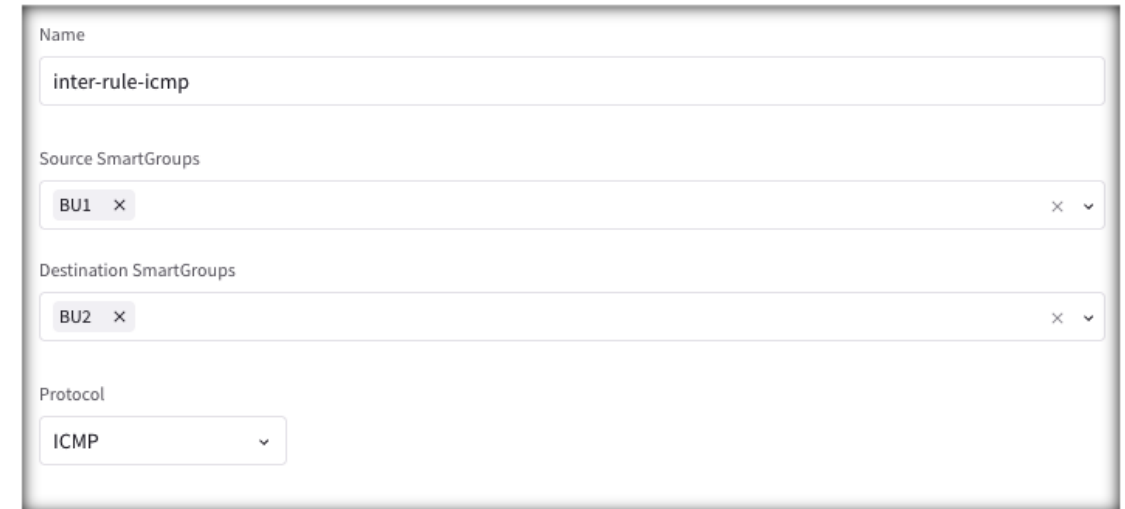
- ❑ Source Smart Group and Destination Smart Group must be the same



The screenshot shows the configuration for an intra-rule. The 'Name' field is 'intra-rule-icmp'. Both the 'Source SmartGroups' and 'Destination SmartGroups' fields contain 'BU1'. The 'Protocol' dropdown is set to 'ICMP'.

2) **Inter-rule** will affect the traffic BETWEEN Smart Groups

- ❑ Source Smart Group and Destination Smart Group must differ



The screenshot shows the configuration for an inter-rule. The 'Name' field is 'inter-rule-icmp'. The 'Source SmartGroups' field contains 'BU1' and the 'Destination SmartGroups' field contains 'BU2'. The 'Protocol' dropdown is set to 'ICMP'.



Next:

Lab 8 Distributed Cloud Firewall